

Transform Validate And Migration Approach(INT to PROD)

Referrals and Appointments

21 August, 2026

Table of Contents

1	Migration Strategy	3
2	FHIR R4 Transformation Process	6
3	Reporting and Validation	8
3.1	Files generated in INT S3 after transformation.....	8
3.1.1	INT bucket - migration folder	8
3.1.2	INT bucket - report folder	8
3.2	Files expected in PROD S3 after migration	8
3.2.1	PROD bucket - migration folder.....	8
3.2.2	PROD bucket - verification/report outputs	9
3.3	Validation expectation	9

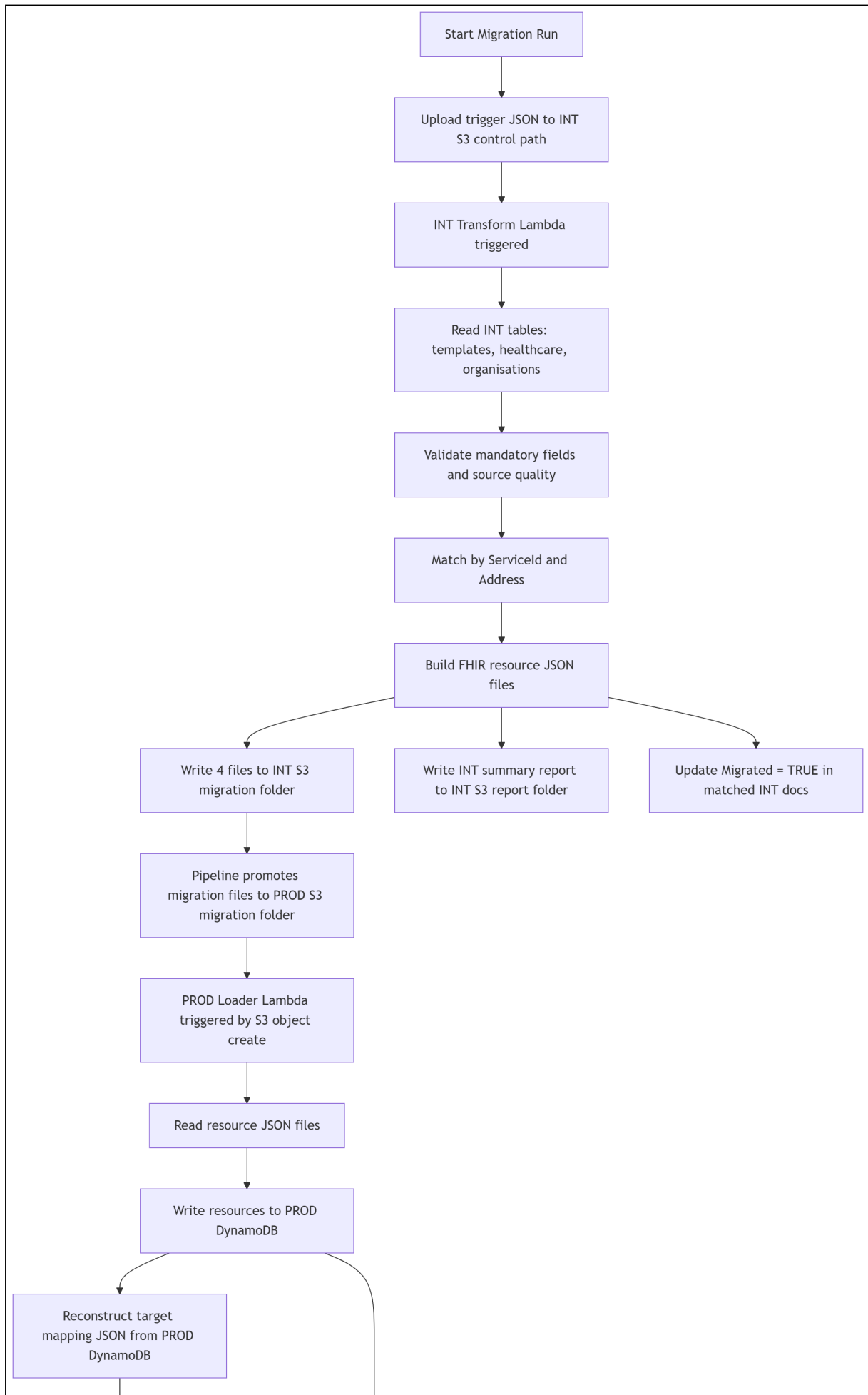
1 Migration Strategy

This migration covers the controlled transformation and promotion of legacy BaRS data from INT into the Endpoint Catalogue in PROD, using FHIR-aligned resource structures and auditable deployment steps. The in-scope source systems are the INT template, healthcare service, and organisation data stores, which are used to generate migration resource JSON files in INT S3. These files are then promoted to PROD S3 through pipeline, where a loader lambda persists resources into PROD DynamoDB and produces verification outputs. The process includes transformation, load, reconciliation, and summary reporting, with explicit counts for matched, transformed, unmatched, and successfully written records. Out of scope for matching is the legacy endpoint table; endpoint resources are derived per matched template rather than reused from historical endpoint rows.

The migration must apply deterministic matching based on ServiceId and canonical address so each healthcare service resolves to the correct template lineage. A single child endpoint is created per template, and each child endpoint must be active with open-ended dates ([period.start = null \(see page 3\)](#), [period.end = null \(see page 3\)](#)). HealthcareService records reference one endpoint, while an endpoint can be referenced by multiple healthcare services. Mandatory FHIR-aligned fields must be present for each resource before write, and only successfully matched records are marked as migrated in INT source documents. The PROD load is considered successful only when DynamoDB write counts match input resource counts and reconstructed target mapping from PROD data matches expected ServiceId-to-address behavior. Any mismatch, write failure, or mapping deviation must be captured in summary output with clear reason codes.

flowchart TD

Referrals and Appointments – Transform Validate And Migration Approach(INT to PROD)



2 FHIR R4 Transformation Process

The transformation process begins when a migration trigger file is uploaded to the INT S3 control location. This invokes the transformation lambda, which reads source data directly from the INT environment rather than from exported CSV files. The source datasets used during transformation are:

- `int_endpointtemplate`
- `int_healthcareservice`
- `int_organisations`

These source records are used to construct the FHIR R4 resources required by the Endpoint Catalogue migration process. The transformation produces four migration output documents:

- Endpoint Template resources
- Endpoint resources
- HealthcareService resources
- Identifier Lookup resources

The transformation logic uses `ServiceId` and canonical endpoint address as the primary matching keys. The legacy endpoint table is not used as a migration source for matching. Instead, once a template match is confirmed, the migration process creates one child Endpoint for each matched template. Each generated child Endpoint is created with:

- `status = active` (see page 3)
- `period.start = null` (see page 3)
- `period.end = null` (see page 3)

This ensures the endpoint is open-ended and avoids introducing invalid or overlapping date ranges from the legacy model.

Each transformed HealthcareService references a single Endpoint, while each Endpoint may be referenced by multiple HealthcareServices. The transformation also creates the parent-child relationship between Endpoint Templates and child Endpoints, and populates all mandatory FHIR-aligned attributes required by the Endpoint Catalogue data model, including identifiers, organisation references, metadata, endpoint references, and provider associations.

The transformation process does not automatically migrate every record from the source environment. Each candidate resource is validated against the Endpoint Catalogue FHIR/Dynamo data model before inclusion in the migration output. Only records that satisfy mandatory field requirements and pass matching validation are included in the approved migration dataset. This ensures that incomplete, invalid, or unmapped legacy data is excluded from rollout and reported separately for investigation.

In addition to generating the migration resource files, the transformation lambda updates the `Migrated` field to `TRUE` for successfully matched HealthcareService and Template source documents in INT. It also generates a summary report containing the number of matched, transformed, unmatched, and failed records for each resource type.

Example transformation output metrics should be reported in a structured summary, such as:

Metric	Count
Source Endpoint Templates	13
FHIR Endpoint Templates Created	13
Source Endpoints	13
FHIR Endpoints Created	13
Source HealthcareServices	10,284
FHIR HealthcareServices Created	10,280
Transformation Failures	4
Transformation Status	PASS

3 Reporting and Validation

The migration process produces a defined set of files in S3 at two stages: after INT transformation and after PROD migration. These files provide the migration payload, audit trail, and verification evidence required to confirm that the rollout has completed successfully.

3.1 Files generated in INT S3 after transformation

After the INT transformation lambda completes, the following files should exist in the INT bucket.

3.1.1 INT bucket - migration folder

These files are the migration resource payloads that will later be promoted to PROD through pipeline.

- migration/templates_<runId>.json
- migration/endpoints_<runId>.json
- migration/healthcareServices_<runId>.json
- migration/identifierLookups_<runId>.json

3.1.2 INT bucket - report folder

These files remain in INT and provide traceability for the transformation step.

- report/transformation_summary_<runId>.json
- report/transformation_errors_<runId>.json (if applicable)
- report/unmatched_records_<runId>.json (if applicable)

3.2 Files expected in PROD S3 after migration

Once the pipeline promotes the migration payload files from INT to PROD, and the PROD loader lambda completes, the following files should exist in the PROD bucket.

3.2.1 PROD bucket - migration folder

These are the promoted input payload files used by the PROD loader lambda.

- migration/templates_<runId>.json
- migration/endpoints_<runId>.json
- migration/healthcareServices_<runId>.json

- `migration/identifierLookups_<runId>.json`

These files should be the same migration resource files generated in INT and copied to PROD via pipeline.

3.2.2 PROD bucket - verification/report outputs

After loading the records into DynamoDB, the PROD lambda should generate verification outputs.

- `report/prod_load_summary_<runId>.json`
- `report/prod_load_errors_<runId>.json` (if applicable)
- `verification/recreated_targets_<runId>.json`

The `recreated_targets_<runId>.json` file is reconstructed from the PROD DynamoDB records and is used to confirm that the final migrated data reproduces the expected service-to-endpoint routing behaviour.

The PROD summary should include:

- number of template records in input file
- number of endpoint records in input file
- number of healthcare service records in input file
- number of identifier lookup records in input file
- number of records successfully written to DynamoDB by type
- number of failed writes by type
- whether recreated target JSON was generated
- whether recreated target JSON was uploaded
- whether recreated target JSON matches expected routing behaviour
- overall PROD migration status

3.3 Validation expectation

The presence of files alone is not sufficient to confirm success. Validation is complete only when:

- all expected migration resource files exist
- PROD summary counts match resource counts in the promoted JSON files
- recreated target JSON is produced successfully
- recreated target JSON reproduces the expected `targets.json` routing behaviour